write the equations that were left out of Eqs. 6.8. Use the appendix to check your answers.

Exercise 6.5: Prove the following theorem:

When any one of Alice's or Bob's spin operators acts on a product state, the result is still a product state.

Show that in a product state, the expectation value of any component of $\vec{\sigma}$ or $\vec{\tau}$ is exactly the same as it would be in the individual single-spin states.

This last exercise proves something important about product states. In a product state, every prediction about Bob's half of the system is exactly the same as it would have been in the corresponding single-spin theory. The same goes for Alice.

An example of this property of product states involves what I called the *Spin-Polarization Principle* in Lecture 3. A useful way to state that principle is:

> *For any state of a single spin, there is some direction for which the spin is* +1.

As I explained, this means that the expectation values of the components satisfy the equation

$$\langle\sigma_x\rangle^2 + \langle\sigma_y\rangle^2 + \langle\sigma_z\rangle^2 = 1, \qquad (6.11)$$

which tells us that not all the expectation values can be zero. This fact continues to hold for all product states. However, it does *not* hold for the entangled state |*sing*>. In fact, for the |*sing*> state the right-hand side of Eq. 6.11 becomes zero, as we'll show next.